

Clinical Study Summary Report (CSSR)

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1. Administrative & Study Identification

Full Study Title: Study of Vibostolimab Alone and in Combination With Pembrolizumab in Advanced Solid Tumors (MK-7684-001) (KEYVIBE-001) **Short Title/Acronym:** KEYVIBE-001 **Protocol Number:** MK-7684-001 **NCT Number:** NCT02964013 **Posting ID:** 1042606673388270708 **Sponsor Name:** Merck **Company:** Merck **Trial Status:** COMPLETED **Investigational Product:** Vibostolimab (MK-7684, anti-TIGIT antibody), Pembrolizumab (MK-3475, anti-PD-1 antibody), Pemetrexed, Carboplatin, Cisplatin (Trade Name: Platinol), and Etoposide. **Phase of Development:** Phase 1 **Medical Monitor:** Clinical Director, Oncology Early Development, Merck & Co., Inc.

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To estimate the recommended Phase 2 dose (RPTD) and evaluate the safety, tolerability, and pharmacokinetics (PK) of vibostolimab (MK-7684) administered as monotherapy or in combination with pembrolizumab (MK-3475), pembrolizumab plus pemetrexed and carboplatin, or pembrolizumab plus platinum agent (carboplatin or cisplatin) and etoposide in adults with metastatic solid tumors. **Secondary Objectives:** To evaluate the anti-tumor activity (including Objective Response Rate [ORR] per RECIST v1.1) of vibostolimab alone and in combination therapies, including specific cohorts such as PD-1 refractory non-small cell lung cancer (NSCLC), PD-1 naive NSCLC, and Japanese participants with gastric cancer.

2.2 Background & Rationale To address the need for therapies in advanced solid tumors where patients do not respond or develop resistance to current immune checkpoint inhibitors. TIGIT is an inhibitory receptor coexpressed with PD-L1 that contributes to T-cell suppression. The addition of the anti-TIGIT antibody vibostolimab to the PD-1 inhibitor pembrolizumab (evaluated both as separate infusions and as a fixed dose coformulation, MK-7684A) is hypothesized to prevent the immunosuppressive effects of TIGIT and provide improved anti-tumor efficacy.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm / Non-randomized (Dose Escalation and Expansion Cohorts - Part A & B), with an embedded 1:1 randomized evaluation in Part B for PD-1 treatment-naive cancers. **Blinding:** Open-label

Randomization: Used in specific expansion cohorts (e.g., Part B for PD-1 treatment naive cancer).

3.2 Planned Enrollment & Duration Target Enrollment: 474 participants **Number of Sites:** Global Multicenter (including specific cohorts from mainland China and Japan) **Estimated Study Start:** 11-Nov-2016 **Status:** Completed

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adults with a histologically or cytologically confirmed metastatic solid tumor for which there is no available therapy that is expected to convey clinical benefit. 2. Measurable disease per Response Evaluation Criteria in Solid Tumors (RECIST) v1.1. 3. Eastern Cooperative Oncology Group (ECOG) performance status of 0 to 1. 4. Part A Japanese cohort: Must reside in Japan, be of Japanese descent, and have inoperable adenocarcinoma of the stomach and/or gastric-esophageal junction (GEJ) that progressed on at least 1 prior therapy. 5. Part A NSCLC cohort: Stage IV non-squamous NSCLC. 6. Part B China cohort: Chinese from mainland China with metastatic solid tumor with no more than 2 prior lines of therapy.

4.2 Exclusion Criteria 1. Receipt of chemotherapy, radiation, biological cancer therapy, or major surgery within 4 weeks prior to the first dose. 2. Active, known, or suspected autoimmune disease or inability to interrupt immunosuppressive NSAIDs/therapies. 3. Previous treatment with another agent targeting the TIGIT receptor. 4. Active central nervous system (CNS) metastases and/or carcinomatous meningitis. 5. Known human immunodeficiency virus (HIV) positive and/or known active chronic or acute Hepatitis B or Hepatitis C. 6. Receipt of a live virus vaccine within 30 days prior to the first dose of study treatment. 7. Receipt of hormonal cancer therapy (e.g., tamoxifen, leuprolide) within 4 weeks prior to the first dose of study treatment.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Vibostolimab at assigned dose levels alone, or in combination with Pembrolizumab, or as a coformulated fixed dose (MK-7684A), administered every 3 weeks. Additional combinations include pembrolizumab plus pemetrexed and carboplatin, or pembrolizumab plus the investigator's choice of platinum agent (carboplatin or cisplatin/Platinol) and etoposide. **Route of Administration:** Intravenous (IV) infusion **Duration of Treatment:** Until documented disease progression, unacceptable toxicity, or completion of maximum cycles.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care medications for disease-related symptoms.

Prohibited: Systemic immunosuppressive therapies, live viral vaccines (within 30 days), concurrent investigational agents, and prior immunomodulatory discontinuation due to Grade 3+ immune-related adverse events.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline Imaging, Tumor Tissue Sample Collection
Baseline	Day 0	First Dose, Safety Labs, PK/PD Sampling
Interim	Every 3 Weeks	Safety Labs, PK Sampling, Investigational Product Administration, Efficacy Assessment
End of Study	+30 Days Post-Dose	Final Vital Signs, Safety Follow-Up, Survival Status Monitoring

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints

- 1. Number of Participants With Dose Limiting Toxicities (DLTs) up to 24 Months:** Evaluated per Common Terminology Criteria for Adverse Events Version 4.0 (CTCAE 4.0).
- 2. Number of Participants Who Experienced At Least One Adverse Event (AE):** Tracking of any untoward medical occurrence temporally associated with study treatment.
- 3. Number of Participants Who Discontinued Study Treatment Due to an AE.**

7.2 Secondary Endpoints

- 1. Overall Response Rate (ORR) in Participants in Vibostolimab Dose Escalation Phase:** Assessed per RECIST Version 1.1.
- 2. ORR in Participants With Programmed Death 1 (PD-1) Refractory Non-small Cell Lung Cancer (NSCLC):** Assessed per RECIST 1.1.
- 3. ORR in Participants With PD-1 Naïve NSCLC:** Assessed per RECIST 1.1.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via clinical assessments and laboratory evaluations, focusing on DLTs and standard oncology toxicity scales (CTCAE v4.0). **Serious Adverse Events (SAEs):** Routine 24-hour reporting timeline for severe/life-threatening events. **Data Monitoring Committee (DMC):** Internal safety monitoring to guide dose-escalation tier progression and expansion cohort initiation.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: All-Participants-As-Treated for the safety set; Efficacy evaluable set for tumor response analysis. **Sample Size Justification:** Empirical justification appropriate for a

Phase 1 multi-cohort dose escalation and subsequent non-randomized and 1:1 randomized expansion cohorts. **Handling of Missing Data:** Participants with missing response evaluations due to early dropout are typically classified as non-responders in ORR analyses.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Standard oncology trial monitoring with frequent verification of source data, focusing on safety, DLTs, and RECIST-based radiographic records. **Audit Trail:** Robust Electronic Data Capture (EDC) systems utilized with data clarification and locking sequences.

11. Study Identifiers & Governance

Primary ID: MK-7684-001 **Registry Number:** NCT02964013 **Posting ID:** 1042606673388270708 **Sponsor/Lead Organization:** Merck **Study Title:** Study of Vibostolimab Alone and in Combination With Pembrolizumab in Advanced Solid Tumors (MK-7684-001) (KEYVIBE-001) **Official Regulatory Title:** A Phase 1 Trial of MK-7684 as Monotherapy and in Combination With Pembrolizumab in Subjects With Advanced Solid Tumors

12. Clinical Framework (Oncology Specific)

Disease Areas: Neoplasms **Primary Condition:** Advanced Metastatic Solid Tumors (including NSCLC and Gastric cancer) **Semantic Type:** Neoplastic Process **Preferred UMLS Name:** Cancer Immunotherapy **Diagnosis Threshold:** Histologically or cytologically confirmed metastatic status **Medical Methodology:** Immune Checkpoint Inhibition (Anti-TIGIT / Anti-PD-1 blockade), Combination Chemotherapy **Optimization Target:** Maximum Tolerated Dose / Recommended Phase 2 Dose (RPTD)

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for Phase 1 dose-finding oncology trials **Education Protocol (HCP):** Identification and management training for immune-mediated adverse events (irAEs) **Education Protocol (Patient):** Trial onboarding, recognition of symptoms of immune toxicities, and adherence tracking **Quality Audit Mechanism:** Clinical response and independent/investigator review for radiologic RECIST assessments

14. Observation & Follow-Up Schedule

Total Duration: Monitored for DLTs up to 24 months; survival follow-up post-treatment. **Onsite Visit Cadence:** Every 3 weeks (Q3W) matching the dosing cycle. **Remote Monitoring Frequency:** Routine safety calls inter-cycle as clinically indicated. **Checklist Review Interval:** Tumor imaging assessments regularly performed per RECIST v1.1.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				